

Preliminary Project Proposal

Reducing Manual Triage Inconsistencies Through Explainable AI-Assisted Triage

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Problem Statement

Triage nurses at low-resource Caribbean health centers sort patients using subjective judgment, producing inconsistent prioritization and elevated mistriage rates (Da'Costa et al., 2025; Tyler et al., 2024). Research shows this method misclassifies up to 1.2% of critical patients, while AI-assisted approaches are nearly 27% more accurate than the average nurse (Tyler et al., 2024). Existing AI triage models were built without clinician input and have not been validated on Caribbean patient data, leaving a critical implementation gap in this context (Araouchi & Adda, 2024; Nsubuga et al., 2025). Thus, I propose a 12-week pilot to build and assess whether a clinician-informed, explainable AI triage prototype reduces triage inconsistency compared to standard nurse-led assessment in a simulated Caribbean ED environment.

Literature Review

Araouchi & Adda (2024), in their research article stated that there was a need for improvement in emergency triaging due to the large number of patients, and complexity of each medical case, overwhelming the nurses who have to triage manually. To come to a solution, the researchers built and tested several AI models designed to predict patient triage levels with precision and accuracy using the Korean Triage and Acuity Scale (KTAS). From this method, the results showed that AI can meaningfully assist in triaging; each model attained a reliable score, however, the Stacking Model proved to be the overall best performer in accuracy, precision, recall and F1-score. Although the method brought such good results, limitations such as a small dataset from only one country took away from those results due to lack of generalization. Furthermore, a gap that these researchers acknowledged was that a larger and more diverse data set is needed in addition to the need for the AI's decisions to be explainable to medical staff.

Da'Costa et al. (2025) found that ED overcrowding, inconsistencies in triaging and the inability of traditional systems to adapt to stressful situations are all creating serious risks to patient safety and staff wellbeing. The authors conducted a narrative review of various peer reviewed studies published between 2014 and 2024 that noted data collection, algorithmic models, real time analytics and natural language processing to assess the benefits, challenges and future directions of AI in triaging. AI triaging systems demonstrated the ability to reduce patient wait times by up to 30%, reduce treatment time by 20%, improve consistency of triage decisions and better manage resources during regular operations and high stress events. The authors flagged a gap in the racial and gender bias in existing AI models as a critical unresolved

problem, alongside the absence of legal frameworks for accountability when AI causes harm and insufficient clinical training.

Tyler et al. (2024) noted that the current nurse led triage process is inconsistent, subjective and under severe pressure from overcrowding, staff shortages and rising patient complexity. To further analyze this issue, the authors conducted a scoping review of 29 peer reviewed US journal studies between 2013 and 2023 from reliable online databases that measured the outcomes of using AI in the triaging process. Across all the studies, AI and ML models consistently outperformed the conventional triaging systems, reduced mistriage rates, improved identification of critically ill patients and, in only one case, surpassed average nurse accuracy by 26.9%. Nearly all studies relied on historical patient data from individual hospitals, making it impossible to confirm how these AI models would actually perform in a live clinical environment across different types of hospitals. The authors also identified gaps including the need for further testing across various hospitals and countries, ethical guidelines around data and patient privacy, and greater clinician involvement in the development and adoption of AI triage systems.

Tahernejad et al (2024) in their review spoke about the use of manual triage and paper-based data collection in hospitals by first responders. AI assisted triage outdates, and even surpasses these processes by being able to track a patient's condition over time, reducing human error in high pressure environments and allows for increased retention of detailed patient information. In conducting this review, the authors sources studies from various medical article databases which focused on applying AI to triage in disaster and emergency settings, rather than

routine hospital conditions. Assessing the outcome, they found that AI assisted triage systems allowed first responders to process approximately three times more patients than traditional methods, while AI diagnostic tools achieved up to 94.57% accuracy in identifying respiratory illnesses during the COVID-19 pandemic. Although the outcomes were impressive, nearly all studies were conducted in a simulated disaster environment, rather than real ones. As a gap, the authors mentioned a lack of AI triage systems due to trust issues, insufficient training, and data privacy concerns.

Nsubuga et al. (2025) found that most low and middle income countries utilize the Kampala Trauma Score (KTS) to triage patients. This method of scoring has been debated against due to its limited predictive accuracy, with patients being scored incorrectly and critical cases being missed entirely. This study took place in rural Uganda and used over 4,000 real patient records to train four machine learning models. While only using basic information from the patient records, all four machine learning models outperformed KTS by a significant margin and were far more reliable at correctly distinguishing patients who needed urgent care from those who could wait. Since the study was conducted in a single hospital in Uganda, the models cannot be generalized across many low resource settings with different numbers of patients, or injuries.

Targeted Gaps

Araouchi & Adda (2024) mentioned that AI's decision making in triage needs to be explainable to medical staff so that clinicians can understand and trust its recommendations. What this means is that most AI triage models work by plainly taking in patient data, evaluating it, and handing out a score in return without explaining its process. The clinician is left contemplating the why factor for numerous cases. The AI's plain score and plain justification is not enough for the triage nurse to feel firm about the decision that is being made.

Similarly, Tyler et al. (2024) stated that clinicians need to be far more involved in the design and adoption of technology and AI tools. Since most technological systems are built by software engineers with minimal experience in other fields, this makes for impressive software on paper, but is frustrating to use by the intended audience. The gap here lies with the people being closest to the solution, the clinicians, being consulted minimally, or even none at all. These two gaps complement each other and can be addressed simultaneously.

By consulting with clinicians, and even other professionals in the medical field, such as hospital administrators, medical researchers, and other health officials, a proper, robust solution to manual triage can be made through the ethical use of AI.

Proposed Solution

Over the next 12 weeks, this project will build and test an application designed to help triage nurses in Caribbean emergency departments make faster, more consistent decisions about which patients need care first. The app will take in basic patient information, such as vital signs and the reason for their visit, and produce a recommended priority level. Importantly, alongside every recommendation, the app will explain in plain language exactly why that score was given, so the nurse always understands the reasoning behind it and feels confident acting on it. To make sure the app actually works for the people who will use it, at least two structured conversations will be held with emergency healthcare professionals during the design process. Their feedback will directly shape how the app looks, what it shows, and how it communicates its recommendations. This is the heart of the project, building something with clinicians, not just for them.

By the end of the 12 weeks, the project will produce a working version of the app tested against existing patient data, a visual mock-up of the interface, and full documentation of the design process and clinical input received.

Workflow Notes and Diagram

The following notes are taken from De Freitas et al. (2020) to determine the workflow of the emergency department from patient arrival to discharge.

1. Patient Arrival Screening

Patients enter the emergency department through the arrival area by self-arrival or ambulance.

The initial nurse screening then occurs immediately. This initial process is done to determine if the patient is admitted into the ED or requires a less immediate version of care.

AI Note #1: In this initial stage tensions are high and nurses are often in a rush to collect information from the paramedics or the patient. As a result, information can be lost quickly, or missed entirely. To reduce, or even avoid this, AI can be used to assist by collecting information. This could lead to faster screening by collecting all necessary information and filtering out the unnecessary details.

2. Screening Decision

The triage nurse performs an initial assessment to either:

- Redirect if their condition doesn't fit the emergency department
- Allow the patient to proceed to registration in the emergency department and the formal triage begins

At this point, screening is the first major decision point in the flow.

3. Registration

In this process the patient is formally admitted into the ED and basic information is collected, like demographics, contact information and identity.

4. Triage

In triage, the nurses collect information such as vital signs, medical history, patient's complaint, and other clinical observations to rank a patient's condition on a scale from 1 - 5.

AI Note #2: Triage depends heavily on the nurse's judgement. However, that judgement could be clouded by various external factors, such as pressure, stress and understaffing. To assist with triage, AI could intake vital signs, symptoms, risk factors and patient history to produce a detailed and explained outcome to the triage nurse for them to make the final call. AI in this case provides assistance by reducing mistriage rates and providing the nurse with a clear, explainable triage score/outcome.

AI Note #3: Similarly, after the patient is triaged, if their case is non-urgent, like a level 4 or 5, they are set to wait. Oftentimes patients can wait a significantly long time before they ever see a nurse again. To reduce that long gap, nurses could use AI as an assistant to reduce losing track of patients that have already been waiting. Whether it's to check in on them again to note stable conditions, or to re-triage them to change their score, AI in this scenario would aid the ED process by a significant margin.

5. Investigations

Tests such as ECGs, urine tests and X-rays are done to investigate the patient's condition further.

This is done to reduce the time taken for processes at a later stage.

6. Streaming

Patients are sent to designated treatment zones. Each zone has dedicated staff, resources and the same types of patients.

7. Clinical Assessment and Diagnostic Testing

The emergency department doctor sees the patient in the assigned stream, takes a history, examines the patient and orders any further investigations that would be required.

AI Note #4: AI can assist doctors in their assessment and diagnosis. Whether it's to gather information from the patient and then suggest tests based on that, or to pick up on information that might have been missed in earlier stages of the workflow, an AI assistant in this case would be more efficient by reducing the waiting time for others to be seen.

8. Treatment

In this stage, the patient is on the road to recovery with treatment. Medication is administered, and the patient is monitored.

9. Disposition Decision

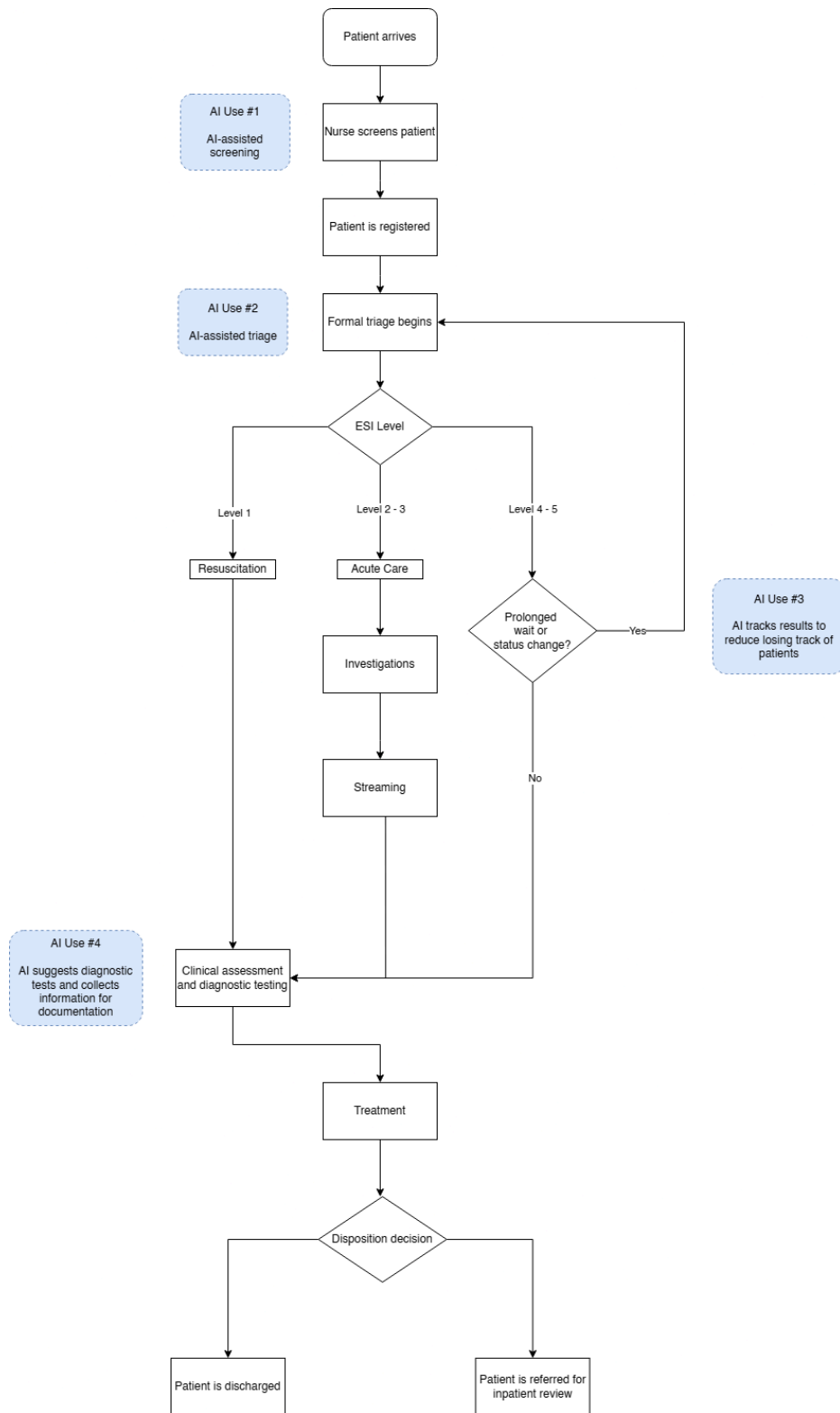
The clinician decides to discharge or do a referral to an inpatient team.

10A. Discharge

If the patient is discharged their treatment is completed, paperwork is finished and they are allowed to leave.

10B. Inpatient Review

If admitted to the inpatient team, the specialists assess the patient in the ED before making a formal admission decision.



Link to full diagram: [ED Flowchart.drawio.png](#)

Constraints

1. Fragmented record system

Mechanism: Clinical information is distributed via paper, travelling with the patient, in addition to verbal handovers, and the nurse's working memory. Therefore, there is no live record that the AI can read.

Design Impact: Any model trained on complete patient records will encounter missing fields constantly which will be treated as a normal case instead of an exception in the real, working world. As a result, confidence scores become unreliable if the model assumes data completeness every time.

Mitigation: Design the model to operate on partial inputs with explicit uncertainty flagging. Each field will be documented to distinguish whether they are required or optional for each task.

2. Past documentation

Mechanism: Some doctors write clinical notes after seeing a patient. As a result the documented state of the patient lags behind the actual clinical state by an unknown variable amount.

Design Impact: The model reads a record that describes the past, instead of the present. Any real time recommendation based on notes may be acting on information that is outdated without even realizing it.

Mitigation: Time stamp all model inferences against the last record update. For example a warning that states “record updated X minutes ago” appears when the information gap exceeds a certain threshold.

3. Connectivity issues

Mechanism: Caribbean public hospitals cannot guarantee stable power, or internet connectivity at all times.

Design Impact: A cloud based model will fail discretely, or even visibly at moments, during high traffic and times when it is most needed.

Mitigation: The model must function offline or degrade gracefully. Core functions such as triage scoring and investigation suggestion should run locally, with sync occurring when connectivity resumes.

Stakeholders

1. Sister Patrice Alleyne – ED Nurse in Charge

Primary concern: This stakeholder's primary concern is to ensure that the flow of the ED remains intact, regardless of the situation. Situations such as staff shortages, high traffic and even hard decision making scenarios may arise, regardless, the nurse in charge has to maintain the orderly flow to reduce patient safety risks.

What success looks like: Sister Patrice gets a real time view of which patients are pending transfer, which results are ready, which streams are understaffed and information streams come with complete information so the briefings take 5 minutes instead of 15.

2. Dr. Kemar Ramsay – ED Registrar

Primary concern: Every order pulls him away from a more complex patient that he's currently managing. This creates a significant lag between each patient that he's actively managing and causes traffic downstream.

What success looks like: Dr. Ramsay receives a prefilled order request that he can approve within seconds, rather than writing in information fields from scratch. As a result, this stops the physical movement of papers, and people, and enforces digital movement of documents.

3. Ms. Tanisha Brown – Triage Nurse

Primary concern: Assigning a triage level on busy shifts where staff is short and tensions are high. If she triages a patient incorrectly, that patient waits in the wrong stream and she carries the consequences.

What success looks like: Ms. Brown has a decision support prompt that flags when her vitals and chief complaint combination historically correlates with a higher triage level, giving her a second opinion without requiring a doctor to be physically present to advise her.

4. Mr. Andre Brooks – Hospital IT Lead

Primary concern: Being handed an AI system that requires stable internet connectivity, a live EHR feed, and ongoing cloud infrastructure in a building where connectivity is shaky and paper records are still the primary documentation method. All while being responsible when it fails.

What success looks like: Mr. Brooks receives a system with documented offline modes, clear data sync behavior, and no dependency on infrastructure that the hospital does not reliably have or support. When all these targets are met, Mr. Brooks can confidently deploy the system without the need to build a new infrastructure first.

5. Mrs. Angela Dunkley – Advocate

Primary concern: Elderly and at-risk patients sitting in the waiting area for hours with no update on where they are in the queue/process, no explanation of why they are waiting and no one to ask because every nurse is tied up.

What success looks like: Mrs. Dunkley can point to a visible plain language status board or patient facing update that tells a waiting patient what they are at and what comes next. Thus, reducing stress and then number of times the staff are interrupted by patients asking for information.

Risk Analysis

<i>Risk Number</i>	<i>Risk Name</i>	<i>Category</i>	<i>Likelihood</i>	<i>Impact</i>	<i>Flag</i>
R1	Training data is skewed toward non-Caribbean patients	AI Technical	High	High	No
R2	The model quietly stops working well overtime	AI Technical	High	High	No
R3	The model gives a wrong answer with high confidence	AI Technical	High	High	Conditional
R4	Nobody can explain why the model made a decision	AI Technical	High	Moderate	No
R5	Clinicians start ignoring all alerts, including real ones	Operational	Moderate	High	No
R6	The AI adds extra work instead of reducing it	Operational	High	Moderate	No
R7	Patients don't know that AI is being used in their care	Ethical	High	Moderate	No
R8	Clinicians stop thinking for themselves and blindly follow the AI	Ethical	High	High	No
R9	The model works poorly for rural and low income patients	Equity	High	High	No
R10	Patients without smartphones or digital records get worse assessments	Equity	Moderate	Moderate	No

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